

OSD Graphic Accelerator Design Specification

Ver 0.2

DM270

Department: Imaging

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HISTORY

Version	Date	Author	Notes
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W1BMP23 (0003:06E2) OFFSET : 0x31	32
W1BMP45 (0003:06E4) OFFSET : 0x32	32
W1BMP67 (0003:06E6) OFFSET : 0x33	32
W1BMP89 (0003:06E8) OFFSET : 0x34	32
W1BMPAB (0003:06EA) OFFSET : 0x35	33
W1BMPCD (0003:06EC) OFFSET : 0x36	33
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1. ABSTRACT

This document explains the specification of On Screen Display(OSD) Graphic Accelerator for DSC27 Digital Still Camera (DSC) chip.

2. REFERENCES

3. GLOSSARY

4. FEATURES(SPECIFICATIONS)

- Polarity Selection of Field Signal
- CbCr reverse
- Resize of VGA to NTSC/PAL
- Back Ground Color Selection
- Programable Color Palette
- Color Palette ROM/RAM Selection
- Two Video Window Display
- One OSD Window and other One OSD or Attribute for OSD-window0 Window
- One rectangular Cursor Window
- All Window Displayed Simultaneously
- Display Priority
((top) >Rect-Cursor>OSD1>OSD0>VDO1>VDO0>Background(Bottom))
- Video Window Zoom (x2,x4,H/V)
- Video Window Field/Frame Selection
- OSD Window Zoom (x2,x4,H/V)
- OSD Window Field/Frame Selection
- Video/OSD Window Enable/Disable Control
- OSD BitMap Width 1,2,4,8-bit Selectable
- Blend with Video and OSD (8-step)
- 16-LUT Register for OSD Window
- Cursor Display with Width and Color Selectable
- SDRAM Address Register for Each Window
- SDRAM Offset Address Register for Each Window
- Display Reference Position Register
- Left-Top Position Register for Each Window and Width/Height Register for Each Window
- Ping Pong Buffer Control with H/W Signal
- Transparency
- 16 bit SDRAM mode

5. GENERAL DESCRIPTION

5.1 Over view

The OSD is video data and OSD data read from SDRAM and blended both signals to output to video encoder. All control and parameter registers are set by built in ARM processor. Setting data is affected by VSYNC signal.

6. BLOCK DIAGRAM

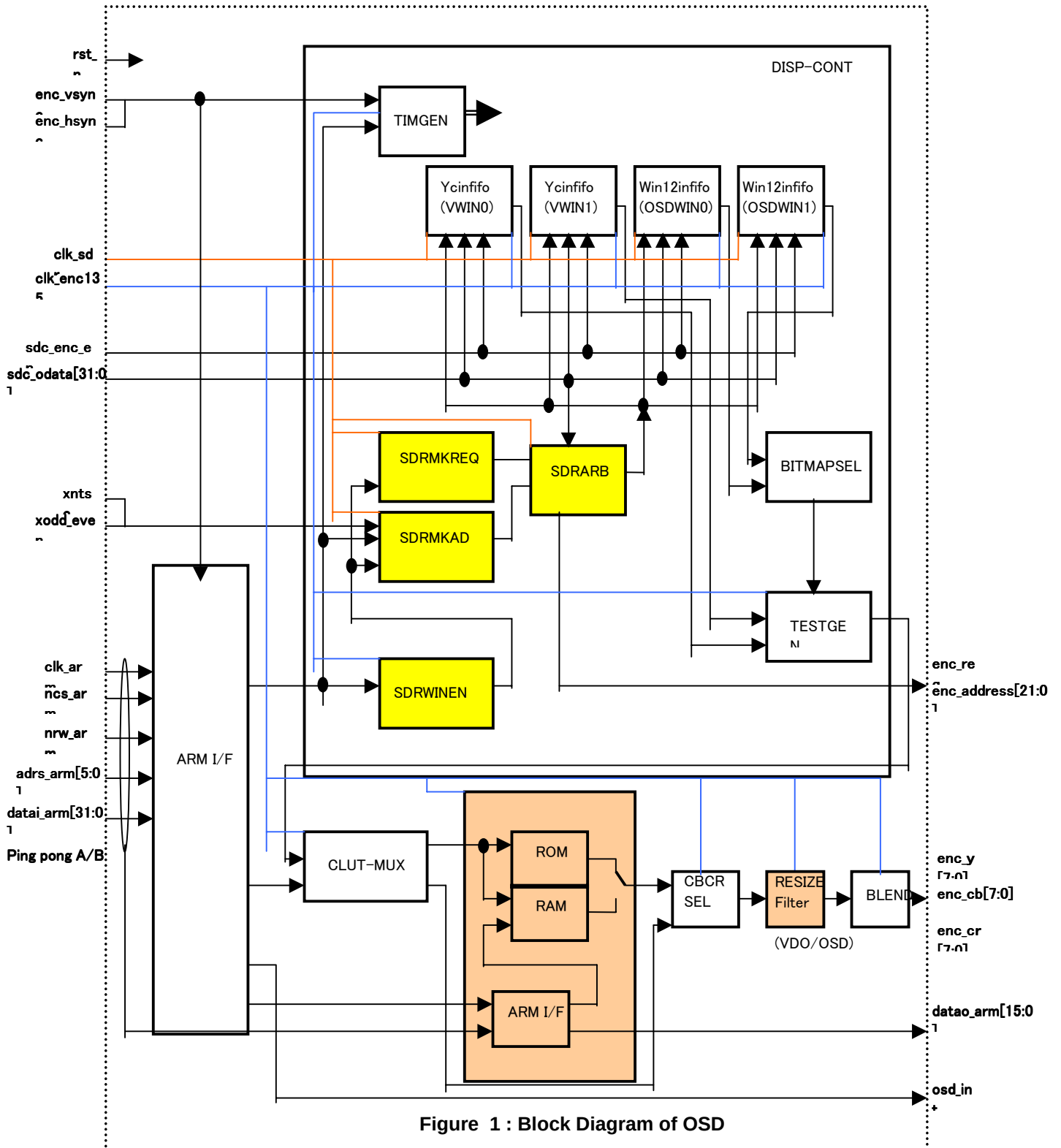


Figure 1 : Block Diagram of OSD

7. Design Files

osd_top.vhd		Design top
_ osd_armif.vhd		ARM interface
_ osd_dispcnt.vhd		Display control top
_ osd_timgen.vhd		Basic timing generator
_ osd_ycinfifo.vhd		FIFO for Video Window-0
_ osd_infifo.vhd		FIFO core
_ osd_ycinfifo.vhd		FIFO for Video Window-1
_ osd_infifo.vhd		FIFO core
_ osd_win12infifo.vhd		FIFO for Bit-Map Window-0
_ osd_infifo.vhd		FIFO core
_ osd_win12infifo.vhd		FIFO for Bit-Map Window-1
_ osd_infifo.vhd		FIFO core
_ osd_sdrarb.vhd		SDRAM Control Arbiter
_ osd_sdrmkad.vhd		SDRAM Address generator
_ osd_sdrmkreq.vhd		SDRAM Data Request Signal Generator
_ osd_sdrwinen.vhd		Window Enable Control
_ osd_testgen.vhd		Internal Test Signal Generator
_ osd_clutmux.vhd		Address Multiplexer for Palette
_ osd_clut.vhd		Color Palette
_ osd_clut_rom.vhd		Color Palette ROM
_ BL00256024040(MB)		Color Palette RAM
_ osd_cbcrsel.vhd		24Bit YCbCr select
_ osd_resize.vhd		Resize filter (for Video)
_ osd_hfilter.vhd		Resize H-filter (for Video)
_ osd_vfilter.vhd		Resize V-filter (for Video)
_ BL00368016040(MB)		V-filterRAM (for Video)
_ osd_blend.vhd		Video/OSD Blend
_ osd_blend_pkg.vhd		Blend Function

Figure 2 : Design files

8. EXPLAIN OF REGISTER and OPERATION

8.1 Relations of windows

The video window is satisfied as follow relations.

Thus,VWIN1 is included in VWIN0. And OSD window0,1 & attribute window include VIDEO window.

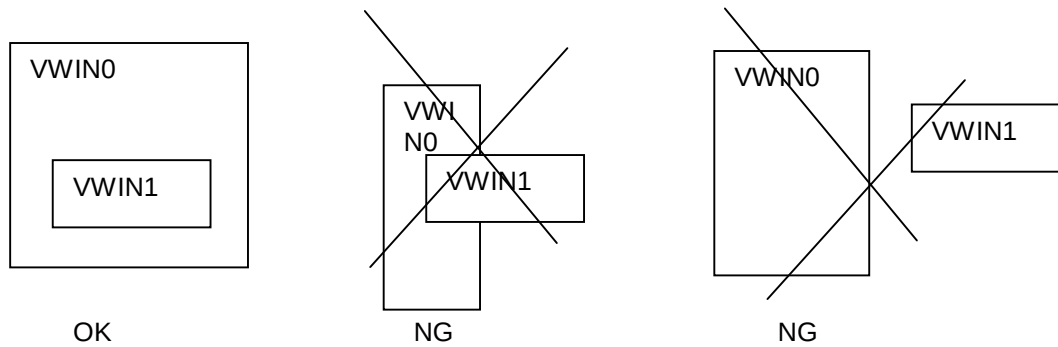


Figure 3 : relations of windows

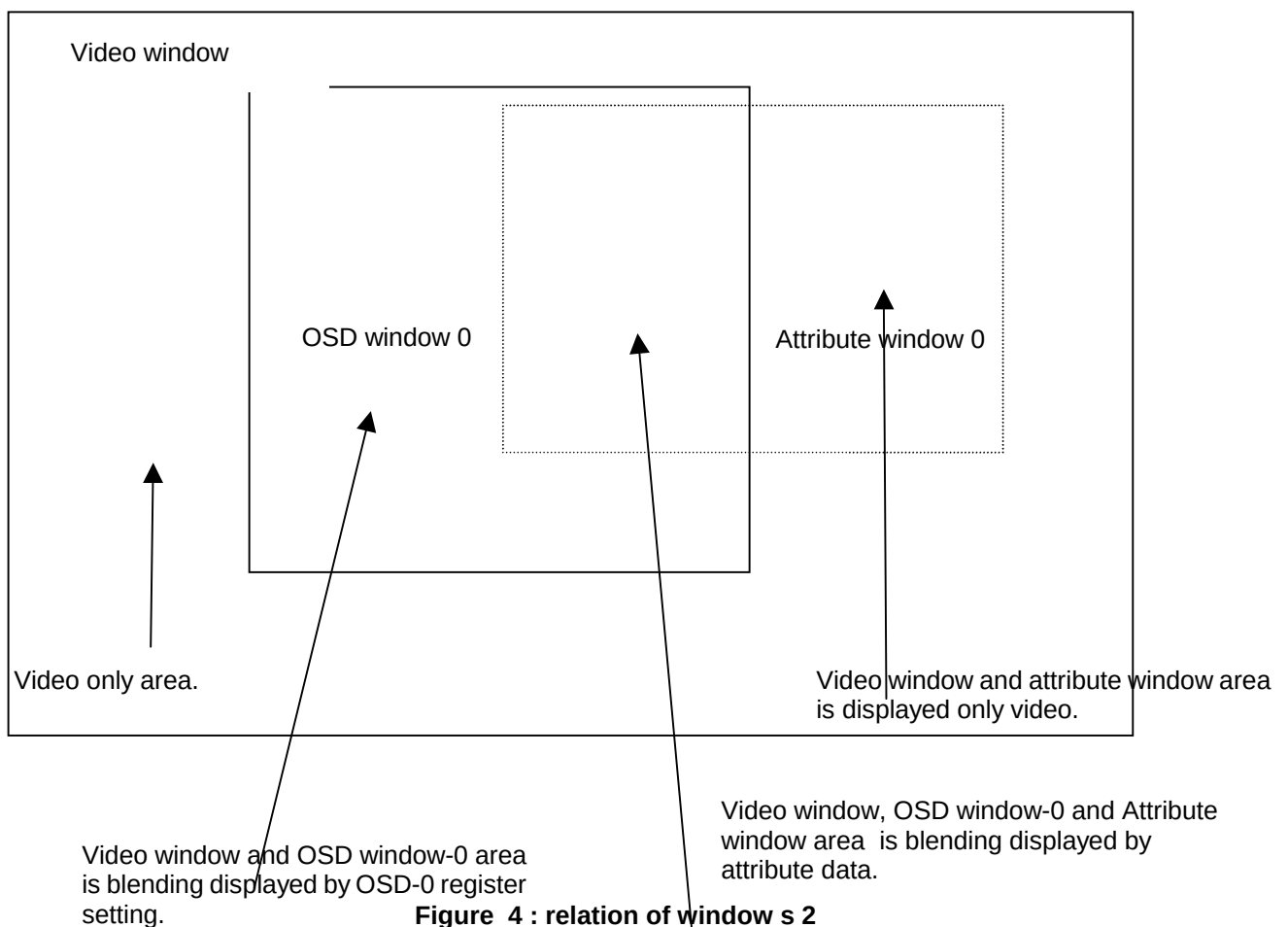


Figure 4 : relation of windows 2

8.2 Relationship between display position

The relationship between display position of each window is shown below.

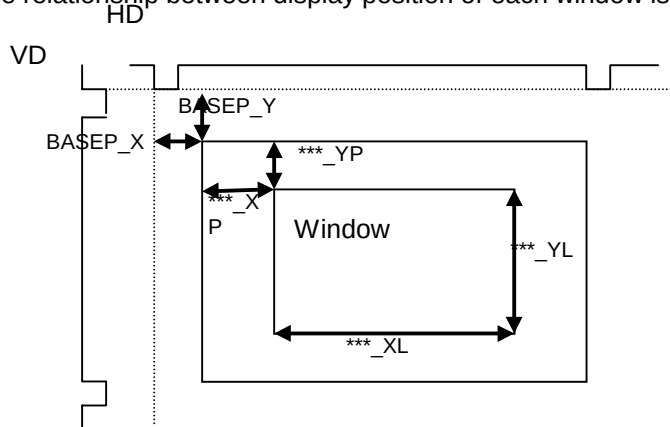


Figure 5 : display position

The display priority of the window is as follows.

((top) Rect-Cursor>OSD1>OSD0>VDO1>VDO0>Background(Bottom))

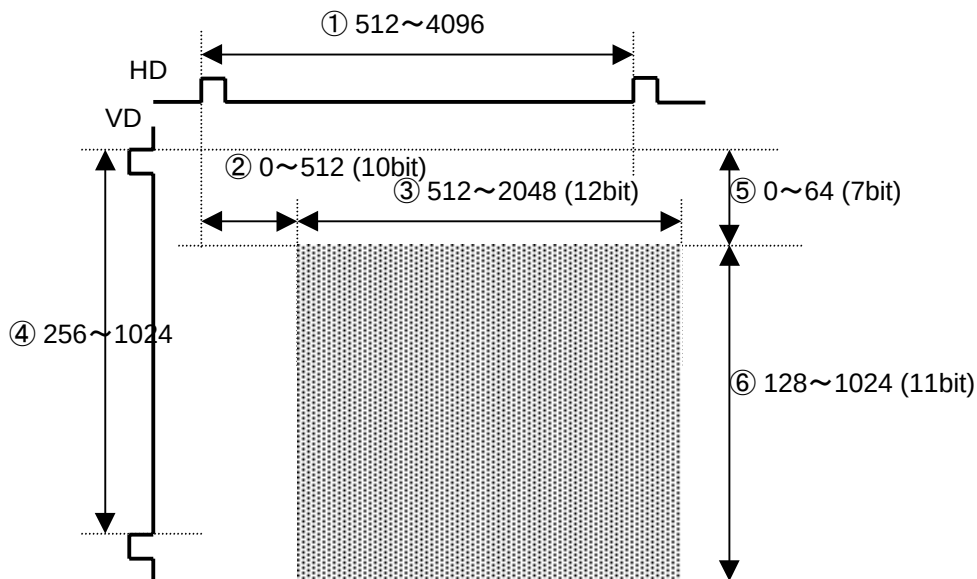


Figure 6 : display position 2

Refer to register of 0x10 to 0x25

8.3 Data format of SDRAM

Picture Data format

Left top

P0	P1	P2	P3	P4	P5	P6	P7...
----	----	----	----	----	----	----	-------

Video Data

Msb	31	24	23	16	15	8	7	0
Px	Y1			Cr0		Y0		Cb0

SDRAM format

Address	msb	lsb
N	P1	P0
N+1	P3	P2
N+2	P5	P4
N+3		

Bit Map Data

8-Bit/pixel

31	24	23	16	15	6	7	0
P3		P2		P1		P0	

SDRAM format

Address	31	23	15	7	0
N	P3	P2	P1	P0	
N+1	P7	P6	P5	P4	
N+2	P11	P10	P9	P8	
N+3					

4-Bit/pixel

31	24	23	16	15	6	7	0
P6	P7	P4	P5	P2	P3	P0	P1

SDRAM format

Address	31	27	23	19	15	11	7	3	0
N	P6	P7	P4	P5	P2	P3	P0	P1	
N+1	P14	P15	P12	P13	P10	P11	P8	P9	
N+2	P22	P23	P20	P21	P18	P19	P16	P17	
N+3									

2-Bit/pixel

31	29	27	25	23	21	19	17	15	13	11	9	7	5	3	1	0
P12	P13	P14	P15	P8	P9	P10	P11	P4	P5	P6	P7	P0	P1	P2	P3	

SDRAM format

Address	31															0
N	P12	P13	P14	P15	P8	P9	P10	P11	P4	P5	P6	P7	P0	P1	P2	P3
N+1	P28	P29	P30	P31	P24	P25	P26	P27	P20	P21	P22	P23	P16	P17	P18	P19
N+2	P44	P45	P46	P47	P40	P41	P42	P43	P36	P37	P38	P39	P32	P33	P34	P35
N+3																

1-Bit/pixel

31																0															
P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
2	2	2	2	2	2	3	3	1	1	1	1	2	2	2	2	8	9	1	1	1	1	1	1	0	1	2	3	4	5	6	7
4	5	6	7	8	9	0	1	6	7	8	9	0	1	2	3			0	1	2	3	4	5								

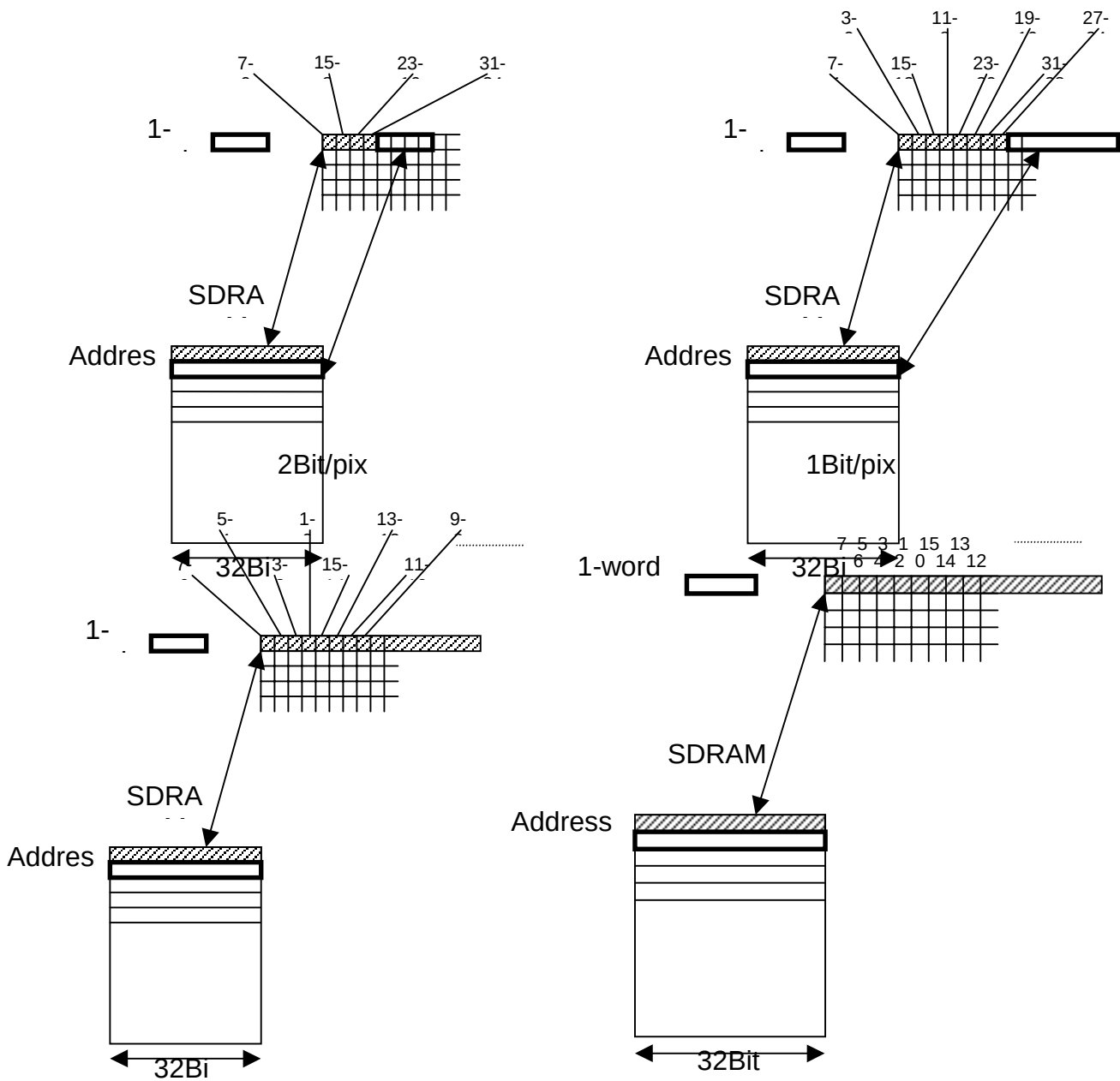
SDRAM format																															
Address 31																0															
N	P24	P25	P26~P3													P4	P5	P6	P7												
N+1	P56	P57	P58~P35													P36	P37	P38	P39												
N+2	P88	P89	P90~P67													P68	P69	P70	P71												
N+3																															

Figure 9 : sdram adress

Bit Map の構成

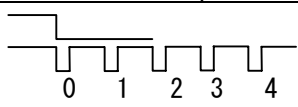
8Bit/pix

4Bit/pix



8.4 Relationship VSYNC and register set

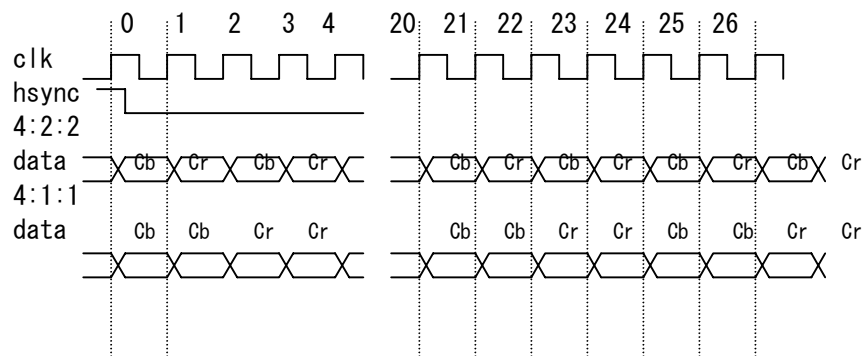
Relationship between VSYNC to picture register set values and position at which data is output

Base_y (0x13)	xxx_yp (0x15, 0x19, 0x1D, 0x21, 0 x25)	Vsync Hsync
		
0	0	1
0	1	2
0	2	3
0	3	4
1	0	1
1	1	2
1	2	3
1	3	4
2	0	2
2	1	3
2	2	4
2	3	5
3	0	3
3	1	4
3	2	5
3	3	6

8.5 Relationship HSYNC and register set

Relationship between HSYNC to picture register set values and position at which data is output

Base_x (0x12)	xxx_xp (0x14, 0x18, 0x0c, 0x20, 0x24)	Input data ← WINDOW START POINT Y: 00 00 1A 14 1B 15 1C 16 Cb: 80 80 1C 1C 1D 1D 1E 1E Cr: 80 80 16 16 17 17 18 18	Numerical value of clock position of timing chart below
22	0	23 24 25 26 27 1A 14 1B 15 1C 80 1C 16 1D 17	23
22	1	24 25 26 27 28 1A 14 1B 15 1C 1C 16 1D 17 1E	24
22	2	24 25 26 27 28 00 1A 14 1B 15 80 80 1C 16 1D	25
22	3	24 25 26 27 28 00 00 1A 14 1B 80 80 1C 16 1D	26
23	0	23 24 25 26 27 1A 14 1B 15 1C 80 1C 16 1D 17	23
23	1	24 25 26 27 28 1A 14 1B 15 1C 1C 16 1D 17 1E	24
23	2	24 25 26 27 28 00 1A 14 1B 15 80 80 1C 16 1D	25
23	3	24 25 26 27 28 00 00 1A 14 1B 80 80 1C 16 1D	26
24	0	24 25 26 27 28 1A 14 1B 15 1C 1C 16 1D 17 1E	24
24	1	24 25 26 27 28 00 1A 14 1B 15 80 80 1C 16 1D	25
24	2	24 25 26 27 28 00 00 1A 14 1B 80 80 1C 16 1D	26
25	0	24 25 26 27 28 00 1A 14 1B 15 80 80 1C 16 1D	25
25	1	24 25 26 27 28 00 00 1A 14 1B 80 80 1C 16 1D	26
25	2	24 25 26 27 28 00 00 00 1A 14 80 80 80 80 1C	27



9. REGISTER MAP

OSD_MODE (0003:0680) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CS	OVRSZ	FSINV	EF	VVRSZ	VHRSZ	RSV	CPS	CABG7	CABG6	CABG5	CABG4	CABG3	CABG2	CABG1	CABG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	CS	Cb/Cr or Cr/Cb format. Select the order of output as Cb/Cr or Cr/Cb 0: Cb/Cr 1: Cr/Cb
14	OVRSZ	Vertical expansion of OSD window. When the video window images are multiplied 6/5 times in the vertical direction, it is sets . It is used when the positive image VGA image is displayed in the PAL television. 0: x1 1: x 6/5
13	FSINV	Field signal reversing bit. 0: Unreversed/not Inversed 1: Reserved/Inversed
12	EF	Expand Filter ON/OFF. ON/OFF of the interpolation filter is specified. It is valid/effective, when VVRSZ or VHRSZ is ON or when the zoom of video window is set. Since the memory for the interpolation filter corresponds only up to the horizontal pixel number 720 pixels, extra care is necessary for using the filter. 0: OFF 1: ON
11	VVRSZ	Vertical expansion of video window. When the video window images are multiplied 6/5 times in the vertical direction, it is sets . It is used when the positive image VGA image is displayed in the PAL television. 0: x1 1: x 6/5
10	VHRSZ	The horizontal expansion of video window. When the video window images are multiplied 9/8 times in the vertical direction, it is sets . It is used when the positive image VGA image is displayed in the NTSC/PAL television. 0: x1 1: x 9/8
9	Resevered	Please set 0 .
8	CPS	Color Palete Selection SW 0: ROM 1: RAM
7-0	CABG7-CABG0	The background color of the image to be display is specified in the address of CLUT. As for the part where the image is not displayed, the color specified in this register is displayed.

VIDEOWIN_MODE (0003:0682) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	VHZ11	VHZ10	VVZ11	VVZ10	VFF1	ACT1	RSV	RSV	VHZ01	VHZ00	VVZ01	VVZ00	VFF0	ACT0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15-14	RSV	Reserve Bit.Do not use.

Bits	Register Name	Description
13-12	VHZ11-VHZ10	Video Window1 Horizontal zoom. VHZ11 VHZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
11-10	VVZ11-VVZ10	Video window 1 vertical zoom VVZ11 VVZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
9	VFF1	Video Window1 display mode. 0: Field mode. Reading every 1 line. 1: Frame Mode. Reading every line.
8	ACT1	Setting the ON/OFF of video window 1 image display. 0: OFF 1: ON
7-6	RSV	Reserve Bit .Do not use.
5-4	VHZ01-VHZ00	Video Window 0 Horizontal Zoom. VHZ11 VHZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
3-2	VVZ01-VVZ00	Video Window 0 vertical zoom. VVZ11 VVZ10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
1	VFF0	Video Window 0 display mode. 0: Field Mode. Reading every (1) line. 1: Frame Mode. Reading Every Line.
0	ACT0	Setting the ON/OFF of Video Window 0 image Display . 0: OFF 1: ON

OSDWIN0_MODE (0003:0684) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	CLUTS0	OVHZ01	OVHZ00	OVVZ01	OVVZ00	BW01	BW00	BF02	BF01	BF00	T0E	OFF0	OACT
R/W	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-13	RSV	Reserve Bit. Do not use.
12	CLUTS0	CLUT selection. The look-up table which OSD window 0 uses is selected. 0: ROM-Lookup table. 1: RAM-Lookup table.
11-10	OVHZ01-OVHZ00	OSD/Window 0 Horizontal zoom. OVHZ01 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')

Bits	Register Name	Description
9-8	OVVZ01-OVVZ00	OSD/Window 0 Vertical Zoom. OVVZ01 OVVZ00 0 0 x1 0 1 x2 1 0 x4 1 1 予約 (Same as '00')
7-6	BW01-BW00	He width of the bit of the bit map OSD/Window 0 is specified. BW11 BW10 Bit width 0 0 1 0 1 2 1 0 4 1 1 8
5-3	BF02-BF00	The mixture ratio of OSD window 0, video window 0, and 1 is set. BF02 BF01 BF00 OSD Window 0 Video Window 0 0 0 0 1 0 0 1 1/8 7/8 0 1 0 2/8 6/8 0 1 1 3/8 5/8 1 0 0 4/8 4/8 1 0 1 5/8 3/8 1 1 0 6/8 2/8 1 1 1 1 0
2	T0E	ON/OFF of transparency of OSD window 0 is set. When set to "1", for the pixel where the bit mapping value is "0" the mixing with colours specified in OSD007-OSD000 of W0BMP01(0003:06D0) and Video Window 0,1 is done according to blending ratio specified in BF02-BF00.
1	OFF0	OSD Window 0 display Mode. 0: Field Mode. Read by jumping /Skipping1 line. 1: Frame Mode. Read every line.
0	OACT	Setting the ON/OFF of image display. 0: OFF 1: ON

OSDWIN1_MODE (0003:0686) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OASW	RSV	RSV	BIT12	BIT11	BIT10	BIT09	BIT08	BIT07	BIT06	BIT05	BIT04	BIT03	BIT02	BIT001	BIT00
R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15	OASW	OSD Window1, Attribute window conversion SW. 0: OSD Window 1 1: Attribute Window

OASW=0(In case of OSD window 1)

Bits	Register Name	Description
14-13	RSV	Reserve Bit .Do not use.
12	BIT12	CLUTSelection. The look-up table which OSD window 1 uses is selected. 0: ROM-Lookup Table 1: RAM-Lookup Table
11-10	BIT11-BIT10	OSD Window 1 horizontal Zoom. BIT11 BIT10 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')

Bits	Register Name	Description
9-8	BIT09-BIT08	OSD Window 1 Vertical Zoom. BIT09 BIT08 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
7-6	BIT07-BIT06	The width of the bit of the bit map of OSD window 1 is specified. BIT07 BIT06 Bit Width 0 0 1 0 1 2 1 0 4 1 1 8
5-3	BIT05-BIT03	The mixture ratio of OSD window 1, video window 0, and 1 is set. BIT05 BIT04 BIT03 OSD Window1 Video Window 0 0 0 0 1 0 0 1 1/8 7/8 0 1 0 2/8 6/8 0 1 1 3/8 5/8 1 0 0 4/8 4/8 1 0 1 5/8 3/8 1 1 0 6/8 2/8 1 1 1 1 0
2	BIT02	ON/OFF of transparency of OSD window 1 is set. When set to "1", for the ipixel where the bit mapping value is "0" the mixing with colours specified in OSD007-OSD000 of W0BMP01(0003:06D0) and Video Window 0,1 is done according to blending ratio specified in BIT05-BIT03
1	BIT01	OSD window 1 Display mode. 0: Field mode. Read by jumping /Skipping1 line. 1: Frame Mode. Read every line.
0	BIT00	Set the ON/OFF of Image display of OSD Window 1. 0: OFF 1: ON

OASW=1(Attribute・ウィンドの場合)

Bits	Register Name	Description
14-13	RSV	Reserve Bit. Do not use.
12	BIT12	Enter 0.
11-10	BIT11-BIT10	Attribute Window horizontal zoom. BIT11 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
9-8	BIT09-BIT08	Attribute Window vertical zoom. BIT09 BIT08 0 0 x1 0 1 x2 1 0 x4 1 1 Reservation (Same as '00')
7-6	BIT07-BIT06	The blinking interval of the Attribute window is specified. Eight times of the VD signal becomes the basic intervals. BIT07 BIT06 0 0 Basic intervalx1 0 1 Basic intervalx2 1 0 Basic intervalx3 1 1 Basic intervalx4
5-2	BIT05-BIT03	Enter 0000.

Bits	Register Name	Description
1	BIT01	Attribute window display mode. 0: Field Mode. Read by jumping /Skipping1 line. 1: Frame Mode. Read every line.
0	BIT00	Set the ON/OFF of blinker. 0: OFF 1: ON

The data format of Attribute window is as follows.

b3	B2	b1	B0
Blink	Blend factor		

Blink: Set the ON/OFF of Blinker for each pixel.

Blend factor: The mixture ratio of the video window for each pixel is set.

RECTANGLE_CURSOR (0003:0688) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CAC7	CAC6	CAC5	CAC4	CAC3	CAC2	CAC1	CAC0	CLUTS	CHW2	CHW1	CHW0	CVW2	CVW1	CVW0	CACT
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	CAC7-CAC0	The color of the cursor is specified in the address of the look-up table.
7	CLUTS	CLUT Selection. The look-up table used is selected. 0: ROM 1: RAM
6-4	CHW2-CHW0	The width of the line in the horizontal direction of the cursor is specified. 1,4,8,12,16,20,24,28 Pixel
3-1	CVW2-CVW0	The width of the line in the vertical direction of the cursor is specified. 1,2,4,6,8,10,12,14Line
0	CACT	Set the ON/OFF of cursor mode. 0: OFF 1: ON

VIDEOWIN0_OFFSET (0003:068C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V0LO8	V0LO7	V0LO6	V0LO5	V0LO4	V0LO3	V0LO2	V0LO1	V0LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit . Do not use.
8-0	V0LO8-V0LO0	1 line offset in video window 0 is specified. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example Horizontally there are 720 pixel(4:2:2) and the offset value of video window 0 is (720x2)/32=45(0x02d)

VIDEOWIN1_OFFSET (0003:069E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V1LO8	V1LO7	V1LO6	V1LO5	V1LO4	V1LO3	V1LO2	V1LO1	V1LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit . Do not use.
8-0	V1LO8-V1LO0	1 line offset in video window 1 is specified. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example Horizontally there are 720 pixel(4:2:2) and the offset value of video window1 is $720/32=45(0x02d)$

OSDWIN0_OFFSET (0003:0690) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O0LO8	O0LO7	O0LO6	O0LO5	O0LO4	O0LO3	O0LO2	O0LO1	O0LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit. Do not use.
8-0	O0LO8-O0LO0	1 line offset in OSD window 0 is specified. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example The offset value of 64 pixel OSD window 0 with a width of eight bits per pixel, horizontally is $(64 \times 8)/32=2(0x0010)$

OSDWIN1_ATTRIBUTE_OFFSET (0003:0692) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	O1LO8	O1LO7	O1LO6	O1LO5	O1LO4	O1LO3	O1LO2	O1LO1	O1LO0
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit.Do not use.
8-0	O1LO8-O1LO0	Specify the 1 line offset of OSD Window 1 and Attribute Window 1. Specify the number which divides the address distance with the following line that is stored in SDRAM by 32. Example The offset value of 64 pixel OSD window 1 with a width of eight bits per pixel, horizontally is $(64 \times 8)/32=2(0x0010)$

VIDEOWIN_AD_HIGER (0003:0694) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	VO21	VO20	VO19	VO18	VO17	VO16	RSV	RSV	V1A21	V1A20	V1A19	V1A18	V1A17	V1A16
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15-14	RSV	Reserve bit.Do not use.

Bits	Register Name	Description
13-8	V0A21-V0A16	The first address of SDRAM of the image data displayed in video window 0 is specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit..
7-6	RSV	Reserve bit.Do not use.
5-0	V1A21-V1A16	The first address of SDRAM of the image data displayed in video window 1 is specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

VIDEOWIN0_AD_LOWER (0003:0696) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	V0A15	V0A14	V0A13	V0A12	V0A11	V0A10	V0A9	V0A8	V0A7	V0A6	V0A5	V0A4	V0A3	V0A2	V0A1	V0A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	V0A15-V0A0	The first address below 16 bits of SDRAM of image data (YCbCr) displayed in video window 0 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

VIDEOWIN1_AD_LOWER (0003:0698) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	V1A15	V1A14	V1A13	V1A12	V1A11	V1A10	V1A9	V1A8	V1A7	V1A6	V1A5	V1A4	V1A3	V1A2	V1A1	V1A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	V1A15-V1A0	The first address below 16 bits of SDRAM of image data (YCbCr) displayed in video window 1 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

OSDWIN_AD_UPPER (0003:069A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	WA021	WA020	WA019	WA018	WA017	WA016	RSV	RSV	WA121	WA120	WA119	WA118	WA117	WA116
R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	0	0	0	0	0	0	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15-14	RSV	Reserve bit .Do not use.
13-8	WA021-WA016	OSD Window 0 address. The first address above 6 bits of SDRAM of image data (YCbCr) displayed in OSD window 0 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.
7-6	RSV	Reserve bit .Do not use.
5-0	WA121-WA116	OSD Window 1 or address of Attribute window. The first address above 6 bits of SDRAM of image data (YCbCr) displayed in OSD window 1 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

OSDWIN0_AD_LOWER (0003:069C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WA015	WA014	WA013	WA012	WA011	WA010	WA009	WA008	WA007	WA006	WA005	WA004	WA003	WA002	WA001	WA000
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	WA015-WA000	OSD Window 0 address. The first address below 16 bits of SDRAM of image data (YCbCr) displayed in OSD window 0 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

OSDWIN1_ATTRIBUTE_AD_LOWER (0003:068E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WA115	WA114	WA113	WA112	WA111	WA110	WA109	WA108	WA107	WA106	WA105	WA104	WA103	WA102	WA101	WA100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	WA115-WA100	OSD Window 1 or address of Attribute window. The first address below 16 bits of SDRAM of image data (YCbCr) displayed in OSD window 1 are specified. 1 address here becomes an offset from the SDRAM top area and is a 32 bytes unit.

BASEP_X (0003:06A0) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	BPX09	BPX08	BPX07	BPX06	BPX05	BPX04	BPX03	BPX02	BPX01	BPX00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-10	RSV	Reserve bit .Do not use.
9-0	BPX09-BPX00	The horizontal display of video window (standard position)0, 1, and OSD window 0 and 1 is set.Specify the number of pixels from the fall of HD. A minimum, set value is 24 pixels.

BASEP_Y (0003:06A2) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BPY08	BPY07	BPY06	BPY05	BPY04	BPY03	BPY02	BPY01	BPY00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit .Do not use.
8-0	BPY09-BPY00	The horizontal display of video window (standard position)0, 1, and OSD window 0 and 1 is set . Specify the number of pixels from the fall of VD. A minimum, set value is 1 line. When it is "0", it becomes 1 line.

VIDEOWIN0_XP (0003:06A4) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	VOX09	VOX08	VOX07	VOX06	VOX05	VOX04	VOX03	VOX02	VOX01	VOX00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-10	RSV	Reserve bit. Do not use.
9-0	V0X09-V0X00	A display of beginning position in the horizontal direction of video window 0 is set. Specify the number of pixels from the displayed standard position.

VIDEOWIN0_YP (0003:06A6) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V0Y08	V0Y07	V0Y06	V0Y05	V0Y04	V0Y03	V0Y02	V0Y01	V0Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit. Do not use.
8-0	W0Y08-W0Y00	A display of beginning position in the vertical direction of video window 0 is set. The value of the line number corresponding to the field is set. Specify the number of lines from the displayed standard position.

VIDEOWIN0_XL (0003:06A8) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	V0W11	V0W10	V0W9	V0W8	V0W7	V0W6	V0W5	V0W4	V0W3	V0W2	V0W1	V0W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserve bit. Do not use.
11-0	V0W11-V0W0	Length of video window 0 in horizontal direction is specified by the number of pixels.

VIDEOWIN0_YL (0003:06AA) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	V0H10	V0H9	V0H8	V0H7	V0H6	V0H5	V0H4	V0H3	V0H2	V0H1	V0H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserve bit. Do not use.
10-0	V0H10-V0H0	The length of video window 0 in the vertical direction is specified by the number of lines. The value of the line number corresponding to the field is set. For instance, if 240 lines are set in NTSC, it becomes the valid display range of vertical.

VIDEOWIN1_XP (0003:06AC) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	V1X09	V1X08	V1X07	V1X06	V1X05	V1X04	V1X03	V1X02	V1X01	V1X00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-10	RSV	Reserve bit. Do not use.
9-0	V1X09-V1X00	A display of beginning position in horizontal direction of video window 1 is set. Specify the number of pixels from the displayed standard position.

VIDEOWIN1_YP (0003:06AE) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V1Y08	V1Y07	V1Y06	V1Y05	V1Y04	V1Y03	V1Y02	V1Y01	V1Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit. Do not use.
8-0	V1Y09-V1Y00	A display of beginning position in vertical direction of video window 1 is set. Specify the number of pixels from the displayed standard position.

VIDEOWIN1_XL (0003:06B0) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	V1W11	V1W10	V1W9	V1W8	V1W7	V1W6	V1W5	V1W4	V1W3	V1W2	V1W1	V1W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserve bit. Do not use.
11-0	V1W11-V1W0	The length of video window 1 in the horizontal direction is specified by the number of pixel.

VIDEOWIN1_YL (0003:06B2) Offset : 0x19

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	V1H10	V1H9	V1H8	V1H7	V1H6	V1H5	V1H4	V1H3	V1H2	V1H1	V1H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserve bit. Do not use.
10-0	V1H10-V1H0	The length of video window 1 in the vertical direction is specified by the number of Lines. The value of the line number corresponding to the field is set. For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

OSDWIN0_XP (0003:06B4) Offset : 0x1A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	W0X09	W0X08	W0X07	W0X06	W0X05	W0X04	W0X03	W0X02	W0X01	W0X00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-10	RSV	Reserve bit. Do not use.
9-0	W0X09-W0X00	Set the display start position in horizontal direction of OSD window 0. Specify the number of pixels from the display standard position.

OSDWIN0_YP (0003:06B6) Offset : 0x1B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	W0Y08	W0Y07	W0Y06	W0Y05	W0Y04	W0Y03	W0Y02	W0Y01	W0Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit. Do not use.
8-0	W0Y08-W0Y00	Set the display start position in vertical direction of OSD window 0. The value of the line number corresponding to the field is set. Specify the number of lines from the display standard position.

OSDWIN0_XL (0003:06B8) Offset : 0x1C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	W0W11	W0W10	W0W9	W0W8	W0W7	W0W6	W0W5	W0W4	W0W3	W0W2	W0W1	W0W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserve bit. Do not use.
11-0	W0W11-W0W0	Length of OSD window 0 in horizontal direction is specified by the number of pixels.

OSDWIN0_YL (0003:06BA) Offset : 0x1D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	W0H10	W0H9	W0H8	W0H7	W0H6	W0H5	W0H4	W0H3	W0H2	W0H1	W0H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserve bit. Do not use.
10-0	W0H10-W0H0	The length of OSD window 0 in the vertical direction is specified by the number of lines. The value of the line number corresponding to the field is set. For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

OSDWIN1_XP (0003:06BC) Offset : 0x1E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	W0X09	W0X08	W0X07	W0X06	W0X05	W0X04	W0X03	W0X02	W0X01	W0X00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-10	RSV	Reserve bit. Do not use.
9-0	W1X09-W1X00	Set the display start position in horizontal direction of OSD window 1 or attribute window. Specify the number of lines from the display standard position.

OSDWIN1_YP (0003:06BE) Offset : 0x1F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	W1Y08	W1Y07	W1Y06	W1Y05	W1Y04	W1Y03	W1Y02	W1Y01	W1Y00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit. Do not use.
8-0	W1Y08-W1Y00	Set the display start position in vertical direction of OSD window 1 or attribute window. The value of the line number corresponding to the field is set. Specify the number of lines from the display standard position.

OSDWIN1_XL (0003:06C0) Offset : 0x20

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	W1W11	W1W10	W1W9	W1W8	W1W7	W1W6	W1W5	W1W4	W1W3	W1W2	W1W1	W1W0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-12	RSV	Reserve bit. Do not use.
11-0	W1W11-W1W0	Length of OSD window 1 or Attribute window in horizontal direction is specified by the number of pixels.

OSDWIN1_YL (0003:06C2) Offset : 0x21

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	W1H10	W1H9	W1H8	W1H7	W1H6	W1H5	W1H4	W1H3	W1H2	W1H1	W1H0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserve bit. Do not use.
10-0	W1H10-W1H0	Length of OSD window 1 or Attribute window in vertical direction is specified by the number of lines. The value of the line number corresponding to the field is set. For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

CUR_XP (0003:06C4) Offset : 0x22

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	CSX09	CSX08	CSX07	CSX06	CSX05	CSX04	CSX03	CSX02	CSX01	CSX00
R/W	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-10	RSV	Reserve bit. Do not use.
9-0	CSX09-CSX00	The display start position in the horizontal direction of the cursor is set. Specify the number of lines from the display standard position.

CUR_YP (0003:06C6) Offset : 0x23

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CSY08	CSY07	CSY06	CSY05	CSY04	CSY03	CSY02	CSY01	CSY00
R/W	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-9	RSV	Reserve bit. Do not use.
8-0	CSY08-CSY00	The display start position in the vertical direction of the cursor is set. Specify the number of lines from the display standard position.

CUR_XL (0003:06C8) Offset : 0x24

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	CSW11	CSW10	CSW9	CSW8	CSCW7	CSW6	CSW5	CSW4	CSW3	CSW2	CSW1	CSW0
R/W	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0

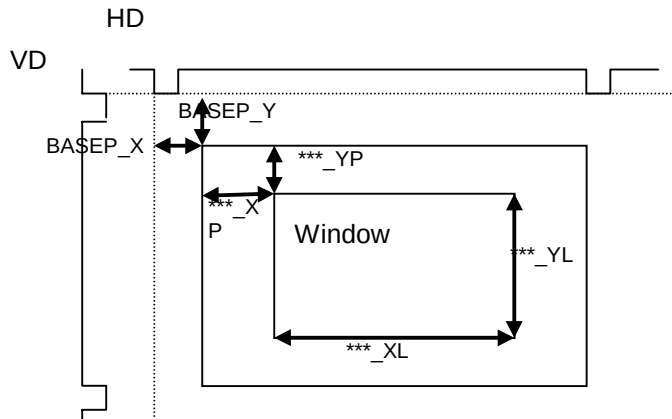
Bits	Register Name	Description
15-12	RSV	Reserve bit. Do not use.
11-0	CSW11-CSW0	Specify the length of the cursor in horizontal direction by number of pixels.

CUR_YL (0003:06CA) Offset : 0x25

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	CSH10	CSH9	CSH8	CSH7	CSH6	CSH5	CSH4	CSH3	CSH2	CSH1	CSH0
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-11	RSV	Reserve bit. Do not use.
10-0	CSH10-CSH0	Specify the length of the cursor in horizontal direction by number of lines. The value of the line number corresponding to the field is set. . For example, if 240 lines are set in NTSC, it becomes the valid/effective range of display for vertical direction.

* the positions of windows as follows.



The display priority of each window is as follows.

Cursor > OSD window 1 > OSDWindow 0 > Video Window 1 > Video Window 0

W0BMP01 (0003:06D0) Offset : 0x28

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD017	OSD016	OSD015	OSD014	OSD013	OSD012	OSD011	OSD010	OSD009	OSD008	OSD007	OSD006	OSD005	OSD004	OSD003	OSD002
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15-8	OSD017-OSD010	The colour of the bit map value "1" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD007-OSD000	The colour of the bit map value "0" is specified in the CLUT(0-255) address .(width=1,2,4)

W0BMP23 (0003:06D2) Offset : 0x29

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD037	OSD036	OSD035	OSD034	OSD033	OSD032	OSD031	OSD030	OSD027	OSD026	OSD025	OSD024	OSD023	OSD022	OSD021	OSD020
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD037-OSD030	The colour of the bit map value "3" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD027-OSD020	The colour of the bit map value "2" is specified in the CLUT(0-255) address . (width=4)

W0BMP45 (0003:06D4) Offset : 0x2A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD057	OSD056	OSD055	OSD054	OSD053	OSD052	OSD051	OSD050	OSD047	OSD046	OSD045	OSD044	OSD043	OSD042	OSD041	OSD040
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD057-OSD050	The colour of the bit map value "5" is specified in the CLUT(0-255) address . (width=2,4)
7-0	OSD047-OSD040	The colour of the bit map value "4" is specified in the CLUT(0-255) address . (width=4)

W0BMP67 (0003:06D6) Offset : 0x2B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD077	OSD076	OSD075	OSD074	OSD073	OSD072	OSD071	OSD070	OSD067	OSD066	OSD065	OSD064	OSD063	OSD062	OSD061	OSD060
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD077-OSD070	The colour of the bit map value "7" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD067-OSD060	The colour of the bit map value "6" is specified in the CLUT(0-255) address . (width=4)

W0BMP89 (0003:06D8) Offset : 0x2C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD097	OSD096	OSD095	OSD094	OSD093	OSD092	OSD091	OSD090	OSD087	OSD086	OSD085	OSD084	OSD083	OSD082	OSD081	OSD080
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
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Bits	Register Name	Description
15-8	OSD097-OSD090	The colour of the bit map value "9" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD087-OSD080	The colour of the bit map value "8" is specified in the CLUT(0-255) address . (width=4)

W0BMPAB (0003:06DA) Offset : 0x2D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD0B7	OSD0B6	OSD0B5	OSD0B4	OSD0B3	OSD0B2	OSD0B1	OSD0B0	OSD0A7	OSD0A6	OSD0A5	OSD0A4	OSD0A3	OSD0A2	OSD0A1	OSD0A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD0B7-OSD0B0	The colour of the bit map value "B" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD0A7-OSD0A0	The colour of the bit map value "A" is specified in the CLUT(0-255) address . (width=2,4)

W0BMPCD (0003:06DC) Offset : 0x2E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD0D7	OSD0D6	OSD0D5	OSD0D4	OSD0D3	OSD0D2	OSD0D1	OSD0D0	OSD0C7	OSD0C6	OSD0C5	OSD0C4	OSD0C3	OSD0C2	OSD0C1	OSD0C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD0D7-OSD0D0	The colour of the bit map value "D" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD0C7-OSD0C0	The colour of the bit map value "C" is specified in the CLUT(0-255) address . (width=4)

W0BMPEF (0003:06DE) Offset : 0x2F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD0F7	OSD0F6	OSD0F5	OSD0F4	OSD0F3	OSD0F2	OSD0F1	OSD0F0	OSD0E7	OSD0E6	OSD0E5	OSD0E4	OSD0E3	OSD0E2	OSD0E1	OSD0E0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD0F7-OSD0F0	The colour of the bit map value "F" is specified in the CLUT(0-255) address . (width=1,2,4)
7-0	OSD0E7-OSD0E0	The colour of the bit map value "E" is specified in the CLUT(0-255) address . (width=4)

W1BMP01 (0003:06E0) Offset : 0x30

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD117	OSD116	OSD115	OSD114	OSD113	OSD112	OSD111	OSD110	OSD107	OSD106	OSD105	OSD104	OSD103	OSD102	OSD101	OSD100
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD117-OSD110	The colour of the bit map value "1" is specified in the CLUT(0-255) address . (width=4)

Bits	Register Name	Description
7-0	OSD107-OSD100	The colour of the bit map value "0" is specified in the CLUT(0-255) address . (width=1,2,4)

W1BMP23 (0003:06E2) Offset : 0x31

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD137	OSD136	OSD135	OSD134	OSD133	OSD132	OSD131	OSD130	OSD127	OSD126	OSD125	OSD124	OSD123	OSD122	OSD121	OSD120
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD137-OSD130	The colour of the bit map value "3" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD127-OSD120	The colour of the bit map value "2" is specified in the CLUT(0-255) address . (width=4)

W1BMP45 (0003:06E4) Offset : 0x32

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD157	OSD156	OSD155	OSD154	OSD153	OSD152	OSD151	OSD150	OSD147	OSD146	OSD145	OSD144	OSD143	OSD142	OSD141	OSD140
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD157-OSD150	The colour of the bit map value "5" is specified in the CLUT(0-255) address . (width=2,4)
7-0	OSD147-OSD140	The colour of the bit map value "4" is specified in the CLUT(0-255) address . (width=4)

W1BMP67 (0003:06E6) Offset : 0x33

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD177	OSD176	OSD175	OSD174	OSD173	OSD172	OSD171	OSD170	OSD167	OSD166	OSD165	OSD164	OSD163	OSD162	OSD161	OSD160
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD177-OSD170	The colour of the bit map value "7" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD167-OSD160	The colour of the bit map value "6" is specified in the CLUT(0-255) address . (width=4)

W1BMP89 (0003:06E8) Offset : 0x34

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD197	OSD196	OSD195	OSD194	OSD193	OSD192	OSD191	OSD190	OSD187	OSD186	OSD185	OSD184	OSD183	OSD182	OSD181	OSD180
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD197-OSD190	The colour of the bit map value "9" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD187-OSD180	The colour of the bit map value "8" is specified in the CLUT(0-255) address . (width=4)

W1BMPAB (0003:06EA) Offset : 0x35

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD1B7	OSD1B6	OSD1B5	OSD1B4	OSD1B3	OSD1B2	OSD1B1	OSD1B0	OSD1A7	OSD1A6	OSD1A5	OSD1A4	OSD1A3	OSD1A2	OSD1A1	OSD1A0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD1B7-OSD1B0	The colour of the bit map value "B" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD1A7-OSD1A0	The colour of the bit map value "A" is specified in the CLUT(0-255) address . (width=2,4)

W1BMPCD (0003:06EC) Offset : 0x36

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD1D7	OSD1D6	OSD1D5	OSD1D4	OSD1D3	OSD1D2	OSD1D1	OSD1D0	OSD1C7	OSD1C6	OSD1C5	OSD1C4	OSD1C3	OSD1C2	OSD1C1	OSD1C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD1D7-OSD1D0	The colour of the bit map value "D" is specified in the CLUT(0-255) address . (width=4)
7-0	OSD1C7-OSD1C0	The colour of the bit map value "C" is specified in the CLUT(0-255) address . (width=4)

W1BMPEF (0003:06EE) Offset : 0x37

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	OSD1F7	OSD1F6	OSD1F5	OSD1F4	OSD1F3	OSD1F2	OSD1F1	OSD1F0	OSD1E7	OSD1E6	OSD1E5	OSD1E4	OSD1E3	OSD1E2	OSD1E1	OSD1E0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	OSD1F7-OSD1F0	The colour of the bit map value "F" is specified in the CLUT(0-255) address . (width=1,2,4)
7-0	OSD1E7-OSD1E0	The colour of the bit map value "E" is specified in the CLUT(0-255) address . (width=4)

CONTROL_MISC (0003:06F4) Offset : 0x3A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PPSW	PPRV	RSV
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	0

Bits	Register Name	Description
15-3	Reseerved	
3	CPBSY	Writing status signal of colour pallete data. 0: Ready 1: Busy
2	PPSW	Change/Switch signal of ping-pong buffer.Referance ()
1	PPRV	PPSW of the above-mentioned register is reversed.
0	RSV	Reserved bit. Set 0.

CLUT_RAM_YCB (0003:06F6) Offset : 0x3B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Y07	Y06	Y05	Y04	Y03	Y02	Y01	Y00	CB07	CB06	CB05	CB04	CB03	CB02	CB01	CB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	Y07- Y00	Writing data (Y) to internal CLUT-RAM. It is necessary to write prior to CLUT_RAM_CR.
7-0	CB07 – CB00	Writing data(Cb) to internal CLUT-RAM. It is necessary to write prior to CLUT_RAM_CR.

CLUT_RAM_CR (0003:06F8) Offset : 0x3C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	CR07	CR06	CR05	CR04	CR03	CR02	CR01	CR00	AD07	AD06	AD05	AD04	AD03	AD02	AD01	AD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-8	CR07- CR00	Writing data(Cb) to internal CLUT-RAM .It is necessary to write continuously in order of CLUT_RAM_YCB and CLUT_RAM_CR.
7-0	AD07 - AD00	CLUT writing RAM address.

VIDEOWIN0_AD0_PING_PONG_BUF (0003:06FC) Offset : 0x3E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	V0PA21	V0PA20	V0PA19	V0PA18	V0PA17	V0PA16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15-6	RSV	Reserved bit. Not used.
5-0	V0PA21-V0PA16	Video window 0 higher/above address of Ping_Pong buffer.

VIDEOWIN0_AD1_PING_PONG_BUF (0003:06FE) Offset : 0x3F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	V0PA15	V0PA14	V0PA13	V0PA12	V0PA11	V0PA10	V0PA09	V0PA08	V0PA07	V0PA06	V0PA05	V0PA04	V0PA03	V0PA02	V0PA01	V0PA00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15-0	V0PA15-V0PA00	Video window 0 Lower/Below address of Ping_Pong buffer.

10. INTERFACE Signal Table

Port Name	Width	In/Out	Description
tst_scan	1	IN	TEST
tst_pmt50	1	IN	TEST
tst_pmt50_ez	1	IN	TEST
tst_pmt50_wz	1	IN	TEST
tst_pmt50_wrenz	16	IN	TEST
tst_pmt50_a	9	IN	TEST
tst_pmt50_d	16	IN	TEST
tst_pmt50_q	16	OUT	TEST
tst_pmt51	1	IN	TEST
tst_pmt51_ez	1	IN	TEST
tst_pmt51_wz	1	IN	TEST
tst_pmt51_wrenz	24	IN	TEST
tst_pmt51_a	8	IN	TEST
tst_pmt51_d	24	IN	TEST
tst_pmt51_q	24	OUT	TEST
tst_pmt52	1	IN	TEST
tst_pmt52_ez	1	IN	TEST
tst_pmt52_wz	1	IN	TEST
tst_pmt52_wrenz	24	IN	TEST
tst_pmt52_a	8	IN	TEST
tst_pmt52_d	24	IN	TEST
tst_pmt52_q	24	OUT	TEST
rst_n	1	IN	Power ON Reset
clk_enc135	1	IN	Video Encoder Clock. It must be synchronized VENC's SCLK output.
clk_sdr	1	IN	SDRAMC Interface Clock
clk_arm	1	IN	ARM Interface Clock
ncs_arm	1	IN	Chip Select of ARM (LOW Active)
nwr_arm	1	IN	Read/Write of ARM (LOW : Read)
adrs_arm	6	IN	Register Address of ARM
datai_arm	16	IN	Write Data Bus of ARM
enc_vsync	1	IN	VSYNC of Video Encoder
enc_hsync	1	IN	HSYNC of Video Encoder
enc_xodd_even	1	IN	Field Selection signal from Video Encoder
sd_c_enc_en	1	IN	Data Valid Identification from SDRAM_C
sd_c_odata	32	IN	Picture Data from SDRAM_C
ping_pongxab	1	IN	Ping-Pong Buffer select signal
sdrc_16mode	1	IN	16 Bit SDRAM mode
enc_req	1	OUT	Data Request to SDRAM_C
enc_address	22	OUT	SDRAMAddress to SDRAM_C

osd_datao_arm	16	OUT	OSD Register Read Data Bus to ARM
osd_nint	1	OUT	VSYNC Interrupt to ARM
enc_y	8	OUT	Luminance signal to Video Encoder
enc_cb	8	OUT	Chrominance(Blue) signal to Video Encoder
enc_cr	8	OUT	Chrominance(Red) signal to Video Encoder